INCH-POUND

A-A-55088
13 September 1991

COMMERCIAL ITEM DESCRIPTION

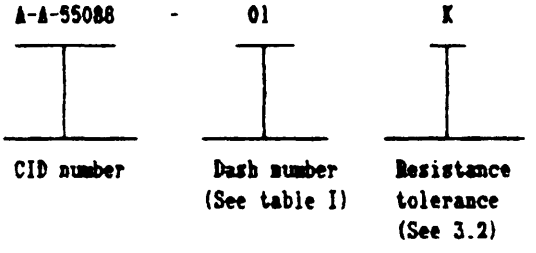
RESISTORS, FLAMEPROOF, FUSIBLE

The General Services Administration
has authorized the use of
this commercial item description.

1. Abstract. This commercial item description (CID) covers the general requirements for flameproof, fusible resistors. These resistors act as a wire-wound resistor and a fuse, and are used in applications where precise control of fusing point and time lags are necessary. These resistors eliminate fire hazard and circuit board damage due to overheated components.

1.1 Application. Resistors covered by this CID are intended for commercial/industrial applications and shall not be used in military systems needing stringent environmental and electrical requirements.

2. Part or identifying number (PIN). The PIN for the CID shall be as shown in the following example:



3. Salient characteristics.

3.1 Resistance value. The resistance range shall be 0.2 ohm minimum to 200 ohms maximum. Specific resistance values are specified in table I by dash number.

3.2 Resistance tolerance. Resistors are available in resistance tolerances (K) ±10 percent, (J) ±5 percent, and (G) ±2 percent.

3.3 Operating temperature. The operating temperature range shall be from -55°C to +150°C.

3.4 Temperature coefficient. The temperature coefficient shall be ±150 ppm < 1Ω, ±50 ppm, 1Ω and above.

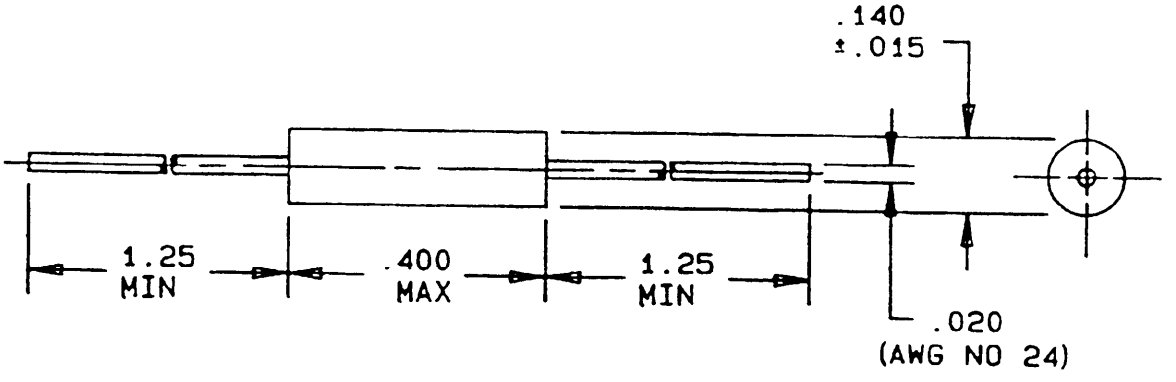
3.5 Fusing characteristics. The fusing characteristics shall be in accordance with table I.

3.6 Design and construction. Resistors supplied to this commercial item description shall be intrinsically safe and shall have a single fusing element, and shall be of an all welded construction with a ceramic bobbin. The configurations and dimensions are shown on figure 1.

3.7 Marking. Resistors supplied to this commercial item description shall be marked with the manufacturers standard commercial PIN.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

inches	mm
.015	.38
.020	.51
.140	3.56
.400	10.16
1.250	31.80

FIGURE 1. Configuration and dimensions.

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TABLE I. Fusing characteristics.

Dash number	Res (Ω)	Steady state		Blow condition I			Blow condition II		
		Current	Dissipation (watts)	Current	Dissipation (watts)	Blow time (T max)	Current	Dissipation (watts)	Blow time (T max)
-01	0.2	2.2 A	1.0	10 A	20	500 ms	25 A	125	50 ms
-02	0.2	2.2 A	1.0	10 A	20	300 ms	25 A	125	30 ms
-03	0.5	1.4 A	1.0	6 A	18	10 sec	15 A	112	100 ms
-04	0.5	1.4 A	1.0	6 A	18	500 ms	15 A	112	50 ms
-05	0.5	1.4 A	1.0	6 A	18	50 ms	15 A	112	10 ms
-06	1.0	1.0 A	1.0	4 A	16	10 sec	10 A	100	100 ms
-07	1.0	1.0 A	1.0	4 A	16	500 ms	10 A	100	50 ms
-08	1.0	1.0 A	1.0	4 A	16	50 ms	10 A	100	10 ms
-09	2.0	700 mA	1.0	3 A	18	10 sec	7 A	98	50 ms
-10	2.0	700 mA	1.0	3 A	18	200 ms	7 A	98	50 ms
-11	2.0	700 mA	1.0	3 A	18	50 ms	7 A	98	10 ms
-12	5.0	450 mA	1.0	2 A	20	200 ms	4.5 A	101	50 ms
-13	5.0	400 mA	0.8	1.5 A	10	50 ms	4.0 A	80	10 ms
-14	5.0	350 mA	0.6	1.0 A	5	100 ms	1.4 A	10	20 ms
-15	10	300 mA	0.9	1.2A	14	200 ms	3.0 A	90	10 ms
-16	10	300 mA	0.9	1.0A	10	100 ms	1.3 A	17	50 ms
-17	10	200 mA	0.4	600 mA	3.6	100 ms	900 mA	8	50 ms
-18	20	220 mA	1.0	600 mA	7.2	10 sec	900 mA	16	100 ms
-19	20	220 mA	1.0	600 mA	7.2	100 ms	900 mA	16	50 ms
-20	20	150 mA	0.4	350 mA	2.4	500 ms	450 mA	4	50 ms

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TABLE I. Fusing characteristics. - continued.

Dash number	RES (Ω)	Steady state		Blow condition I			Blow condition II		
		Current	Dissipation (Watts)	Current	Dissipation (Watts)	Blow time (T Max)	Current	Dissipation (watts)	Blow time (T Max)
-21	50	140 mA	1.0	400 mA	8.0	500 ms	600 mA	18	50 ms
-22	50	130 mA	0.8	350 mA	6.1	100 ms	450 mA	10	50 ms
-23	50	80 mA	0.3	200 mA	2.0	200 ms	240 mA	2.9	50 ms
-24	100	100 mA	1.0	250 mA	6.2	10 sec	300 mA	9.0	100 ms
-25	100	80 mA	0.6	200 mA	4.0	200 ms	250 mA	6.2	250 ms
-26	100	60 mA	0.4	140 mA	2.6	10 sec	180 mA	3.2	200 ms
-27	200	70 mA	1.0	200 mA	8.0	10 sec	250 mA	12.5	100 ms
-28	200	60 mA	0.8	150 mA	4.0	100 ms	250 mA	12.5	10 ms

4. Quality assurance provisions.

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection, examination, and test requirements specified in the contract or purchase order. The contractor may use his own or any other facilities suitable for inspection requirements specified in the contract or purchase order, unless otherwise disapproved by the procuring activity. The acquiring activity reserves the right to perform any of the inspections set forth in the contract or purchase order where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Contractor certification statement. The contractor shall certify and maintain objective quality evidence that the product offered meets the requirements of this CID, and that the product conforms to the producers own drawings, specifications, standards, quality assurances practices, and is the same as the product provided as a bid sample. The acquiring activity reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.3 Certificate of compliance. A certificate of compliance shall accompany all resistors supplied to this CID.

5. Packaging.

5.1 Packaging. Preservation, packaging, labeling, and marking shall be as specified in the contract or purchase order.

6. Notes. This section contains relevant information which is useful to buyers, users, and suppliers in the process of acquiring the item, but is not mandatory.

6.1 Ordering data. The contract or purchase order shall specify the following:

a. CID document number and revision and CID PIN.

b. Quality assurance provisions.

c. Packaging requirements.

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6.2 Suggested sources of supply. A suggested source of supply is listed herein. Additional sources will be added as they become available.

Suggested sources of supply.

CID dash number (see table I)	Vendor commercial PIN	Vendor CAGE number	CID dash number (see table I)	Vendor commercial PIN	Vendor CAGE number
A-A-55088-01	F501	54294	A-A-55088-15	F515	54294
A-A-55088-02	F502	54294	A-A-55088-16	F516	54294
A-A-55088-03	F503	54294	A-A-55088-17	F517	54294
A-A-55088-04	F504	54294	A-A-55088-18	F518	54294
A-A-55088-05	F505	54294	A-A-55088-19	F519	54294
A-A-55088-06	F506	54294	A-A-55088-20	F520	54294
A-A-55088-07	F507	54294	A-A-55088-21	F521	54294
A-A-55088-08	F508	54294	A-A-55088-22	F522	54294
A-A-55088-09	F509	54294	A-A-55088-23	F523	54294
A-A-55088-10	F510	54294	A-A-55088-24	F524	54294
A-A-55088-11	F511	54294	A-A-55088-25	F525	54294
A-A-55088-12	F512	54294	A-A-55088-26	F526	54294
A-A-55088-13	F513	54294	A-A-55088-27	F527	54294
A-A-55088-14	F514	54294	A-A-55088-28	F528	54294

Vendor CAGE number	Vendor name and address
54294	Shallcross US 70 East Smithfield, NC 27577

CONCLUDING MATERIAL

Civil agency coordinating activity:
GSA - 7FXE

Preparing activity:
DLA-ES

(Project 5905-D320)

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